

Submission MICRO1-MID-SUB01

Question MICRO1-MID-Q01

scratch: show step

1. $55 - 3P = 3 + P \Rightarrow P^* = (55 - 3) / (3 + 1) = 13.00$

2. $Q^* = 55 - 3(13.00) = 16.00$

3. substitution gives $Q_d = Q_s$

=> FINAL: $P^* = 13.00$, $Q^* = 16.00$ (check units/interpretation)

Question MICRO1-MID-Q02

scratch: show step

1. $\% \Delta Q = -6/57.0$

2. $\% \Delta P = 0.0952$

3. $E = -1.11$; $|E| > 1$ (elastic)

=> FINAL: elasticity = -1.11; elastic (check units/interpretation)

Question MICRO1-MID-Q03

scratch: show step

1. Shift the wedge between buyer and seller prices by t.

2. Quantity falls relative to equilibrium.

3. The lost mutually beneficial trades create a positive deadweight-loss triangle.

=> FINAL: buyer price minus seller price equals tax; quantity falls; deadweight loss increases (check units/interpretation)

Question MICRO1-MID-Q04

scratch: show step

1. $SMC = PMC + \text{marginal external cost}$, so SMC lies above PMC.

2. A Pigouvian tax equal to marginal external cost aligns private and social incentives.

=> FINAL: social MC above private MC; Pigouvian tax can internalize the external cost (check units/interpretation)

Question MICRO1-MID-Q05

scratch: show step

1. At the ceiling, Q_d exceeds Q_s .

2. Shortage = $Q_d - Q_s$; queues or rationing allocate the scarce quantity.

=> FINAL: $Q_d > Q_s$ creates a shortage; queues/rationing may allocate units (check units/interpretation)

Question MICRO1-MID-Q06

scratch: show step

1. $44 - 4P = 4 + P \Rightarrow P^* = (44 - 4) / (4 + 1) = 8.00$

2. $Q^* = 44 - 4(8.00) = 12.00$

3. substitution gives $Q_d = Q_s$

=> FINAL: $P^* = 8.00$, $Q^* = 12.00$ (check units/interpretation)

Question MICRO1-MID-Q07

scratch: show step

1. $\% \Delta Q = -5/47.5$

2. $\% \Delta P = 0.0952$

3. $E = -1.11$; $|E| > 1$ (elastic)

=> FINAL: elasticity = -1.11; elastic (check units/interpretation)

Question MICRO1-MID-Q08

scratch: show step

1. Shift the wedge between buyer and seller prices by t .

2. Quantity falls relative to equilibrium.

3. The lost mutually beneficial trades create a positive deadweight-loss triangle.

=> FINAL: buyer price minus seller price equals tax; quantity falls; deadweight loss increases (check units/interpretation)